

Figure 13.22: Factors affecting Photosynthesis

5. Water

Photolysis of water provides electrons and protons for the reduction of NADP, directly. Indirect roles are stomatal movement and hydration of protoplasm. During water stress, supply of NADPH + H^+ is affected.

6. Minerals

Deficiency of certain minerals affect photosynthesis e.g. mineral involved in the synthesis of chlorophyll (Mg, Fe and N), Phosphorylation reactions (P), Photolysis of water (Mn and Cl), formation of plastocyanin (Cu).

7. Air pollutants

Pollutants like SO_2, NO_2, O_3 (Ozone) and Smog affects rate of photosynthesis.

13.15.2 Internal Factors

1. Photosynthetic Pigments

It is an essential factor and even a small quantity is enough to carry out photosynthesis.

2. Protoplasmic factor

Hydrated protoplasm is essential for photosynthesis. It also includes enzymes responsible for Photosynthesis.

3. Accumulation of Carbohydrates

Photosynthetic end products like carbohydrates are accumulated in cells and if translocation of carbohydrates is slow then this will affect the rate of photosynthesis.

4. Anatomy of leaf

Thickness of cuticle and epidermis, distribution of stomata, presence or absence of Kranz anatomy and relative proportion of photosynthetic cells affect photosynthesis.

5. Hormones

Hormones like gibberellins and cytokinin increase the rate of photosynthesis.

13.16 Photosynthesis in bacteria

Though we study about bacterial photosynthesis as the last part, bacterial photosynthesis formed first and foremost in evolution. Bacteria does not have specialized structures like chloroplast. It has a simple type of photosynthetic apparatus called chlorosomes and chromatophores (Table 13.6). **Van Neil** (1930) discovered a bacterium that releases sulphur instead of oxygen during photosynthesis. Here, electron donor is hydrogen sulphide (H_2S) and only one photosystem is involved (PS I) and the reaction centre is P_{870}. Pigments present in bacteria are bacteriochlorophyll a, b, c, d, e and g and carotenoids. Photosynthetic bacteria are classified into three groups:

1. Green sulphur bacteria. Example: *Chlorobacterium* and *Chlorobium.*

Table 13.6: Difference between photosynthesis in plants and photosynthesis in bacteria

Photosynthesis in Plants	Photosynthesis in Bacteria
1. Cyclic and non-cyclic phosphorylation takes place	1. Only cyclic phosphorylation takes place
2. Photosystem I and II involved	2. Photosystem I only involved
3. Electron donor is water	3. Electron donor is H_2S
4. Oxygen is evolved	4. Oxygen is not evolved
5. Reaction centres are P700 and P680	5. Reaction centre is P_{870}
6. Reducing agent is NADPH + H^+	6. Reducing agent is NADH + H^+
7. PAR is 400 to 700 nm	7. PAR is above 700 nm
8. Chlorophyll, carotenoid and xanthophyll	8. Bacterio chlorophyll and bacterio viridin
9. Photosynthetic apparatus – chloroplast	9. It is chlorosomes and chromatophores

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